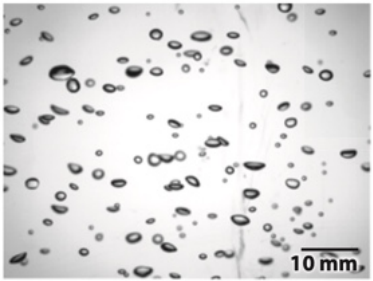
Direct numerical simulations of surfactants in bubbly turbulent flows

74th Annual Meeting of the APS Division of Fluid Dynamics,
2021 November 23

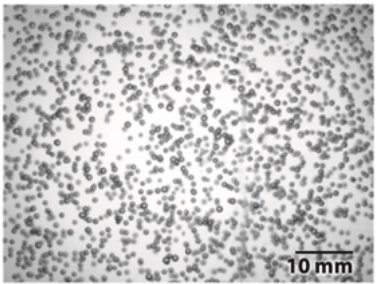
Ianto Cannon, Giovanni Soligo, and Marco E. Rosti,
Complex Fluids and Flows unit, Okinawa Institute of Science and Technology



"The **coalescence** of small bubbles hardly occurs in the presence of surfactant"[1].



Bubbles in tap water.

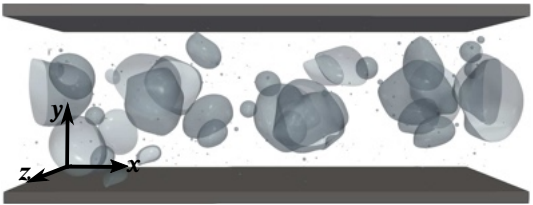


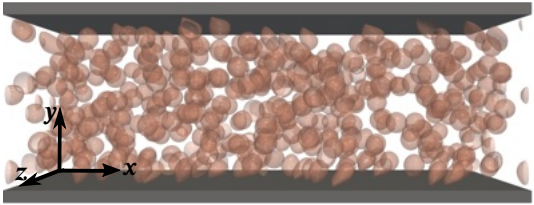
Bubbles in tap water with 2ppm Triton X-100 solution.

[1] Takagi & Matsumoto 2011 *Annual Review of Fluid Mechanics* 43, 615–636

Geometry 1: Bubbles in channel flow

Friction Reynolds number	$Re_\tau \equiv u_\tau L/\nu = 180$
Four volume fractions	$\Phi = 0\%,\ 2.5\%,\ 5\%,\ 10\%,$
Two values of surface tension	$We_\tau \equiv \rho_c u_\tau^2 L/\gamma = 0.57,\ 1.14$





Bubbles are modelled using the **volume of fluid** (VOF) method[2], allowing **easy coalescence and breakup**.

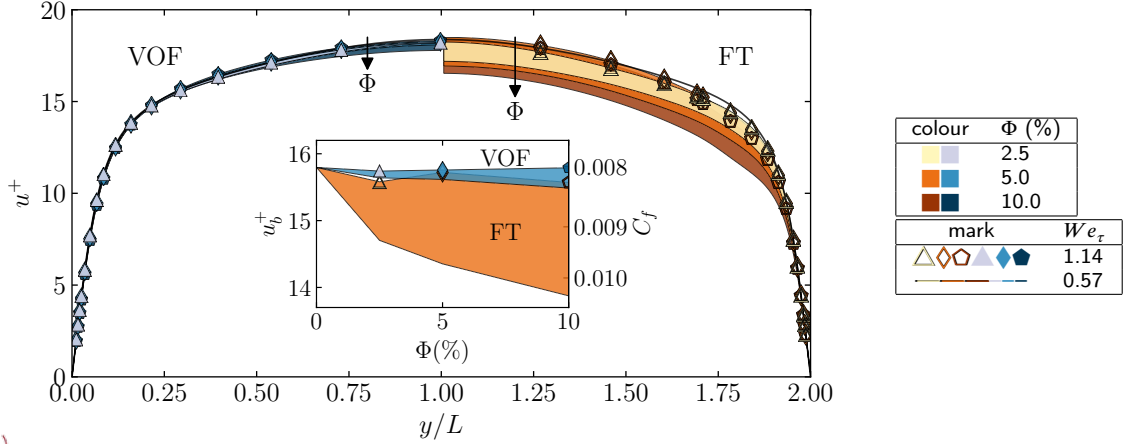
[2] Unverdi & Tryggvason 1992 *Journal of Computational Physics* 100, 25–37.

Bubble interfaces are modelled using the **front tracking** (FT) method[3]. **No coalescence or breakup** occurs.

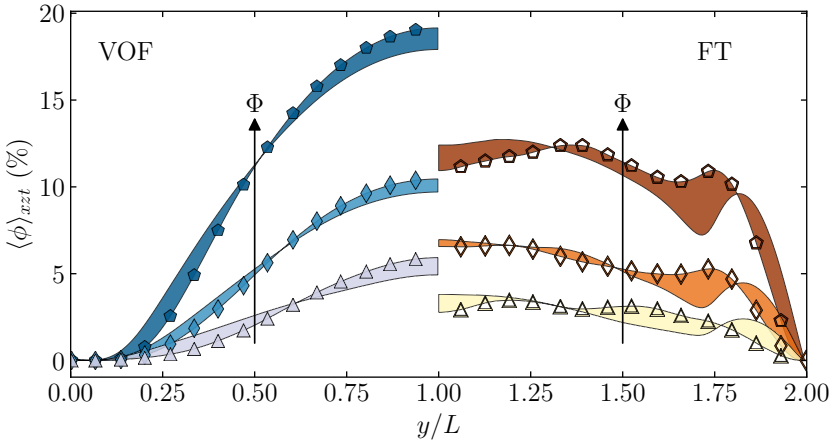
[3] Ii, Sugiyama, Takeuchi, Takagi, Matsumoto & Xiao 2012 *Journal of Computational Physics* 231, 2328–2358.

Flow profile

As the volume fraction of bubbles increases, the **flow rate is reduced**.
Non-coalescing bubbles have a much greater effect.

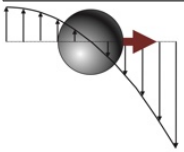


Volume fraction of dispersed phase



colour	Φ (%)
	2.5
	5.0
	10.0

mark	We_τ
	1.14
	0.57



Martel & Toner 2014 Ann. Rev. of Bio. Eng. 16, 371–396

Volume of fluid bubbles are able to coalesce and move to a region of low shear[4].

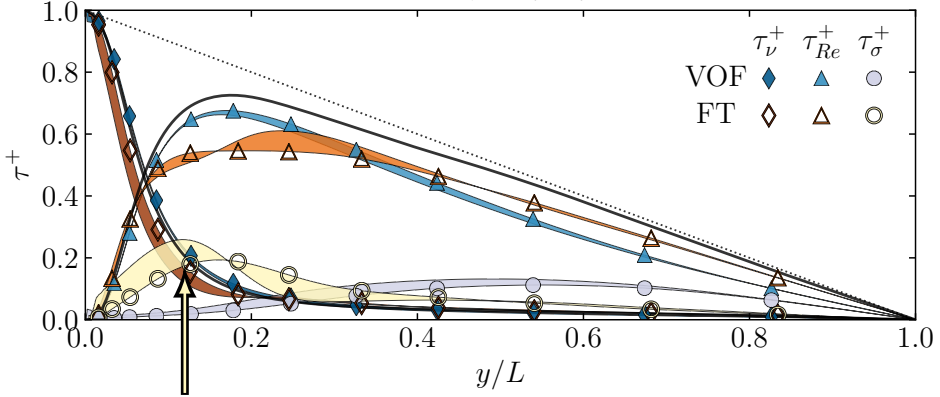
Front tracking bubbles experience a **shear-gradient lift force**, pulling them toward the wall.

[4] Alghalibi, Rosti & Brandt 2019 Phy. Rev. Fluids, 4(10):104201

Shear stresses

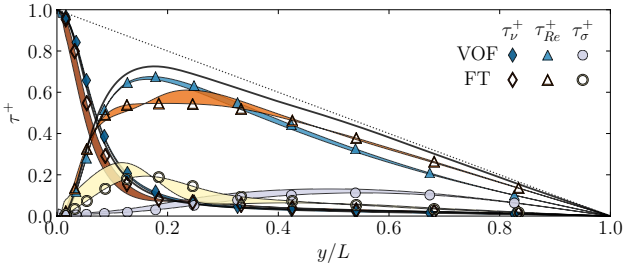
The shear stress is a sum of the viscous, inertial, and surface tension parts,

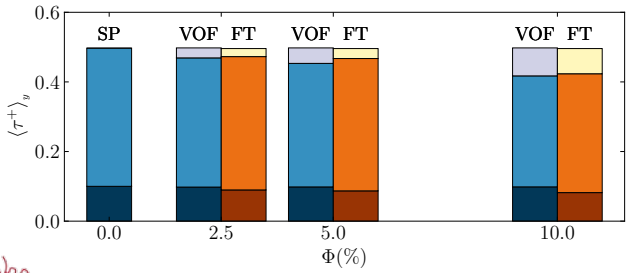
$$1 - y/L = \tau_\nu^+ + \tau_{Re}^+ + \tau_\sigma^+ \tag{1}$$



The front tracking **surface tension stress** has a peak near the wall.

Shear stresses





The mean viscous shear stress is reduced in the front tracking simulations.
This stress is proportional to the centreline velocity,

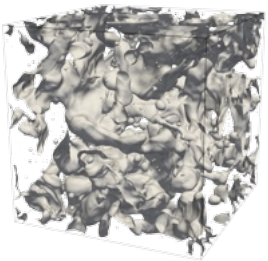
$$\langle \tau_\nu^+ \rangle_y = \frac{1}{L} \int_0^L \frac{du^+}{dy^+} dy = \frac{\delta_\nu}{L} u_{cen}^+. \quad (2)$$

Hence the flow velocity is reduced in the front tracking simulations.

Geometry 2: Bubbles in tri-periodic flow

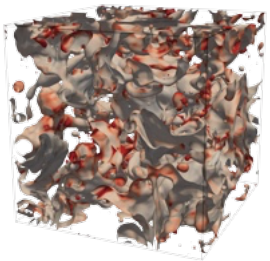
Flow is sustained by stochastic homogeneous isotropic forcing at the large scales[4].
Taylor Reynolds number $Re_\lambda \approx 190$
Surfactant modelled as a third fluid phase, which modifies the surface tension according to the Szyszkowski equation[5]

Clean bubbles



$\phi = 25\%$
$\psi = 0\%$

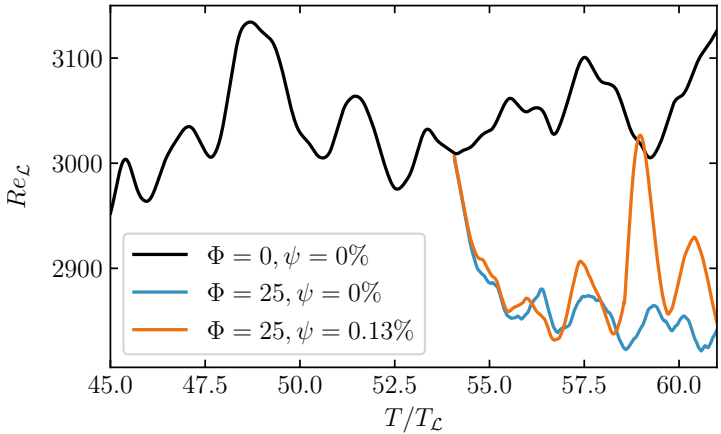
Surfactant-laden bubbles

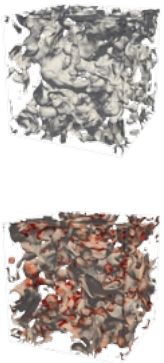


$\phi = 25\%$
$\psi = 13\%$

[4] Eswaran & Pope 1988 *Computers & Fluids*, 16, no. 3, pp. 257–278
[5] Soligo, Roccon and Soldati 2019 *Journal of Computational Physics*, vol. 376, pp. 1292–1311.

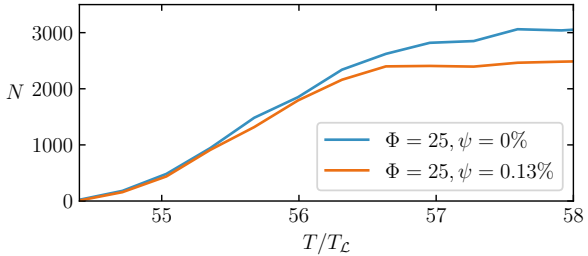
Turbulence at the integral length scales



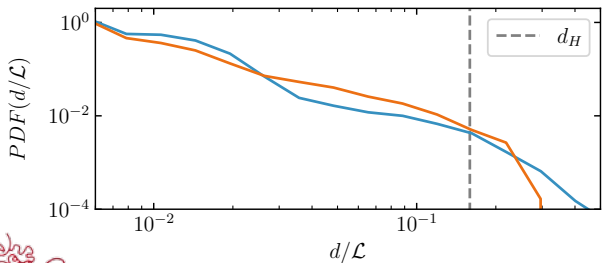


The integral scale Reynolds number $Re_{\mathcal{L}} \equiv U_{RMS}\mathcal{L}/\nu$, is suppressed by the addition of bubbles.

Bubble sizes



N = total number of bubbles.
$T_{\mathcal{L}}$ = integral timescale

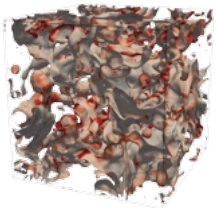


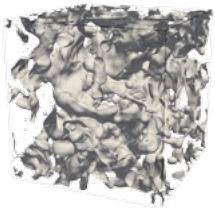
d = droplet diameter
d_H = Hinze scale[4]

The probability density function (PDF) is narrowed by the presence of surfactant.

[4] Hinze 1955 *AIChE J.* vol. 1, no. 3, pp. 289–295.

Conclusions & future work





1. Non-coalescing bubbles produce a significant **drag increase** compared to coalescing bubbles.
2. The non-coalescing bubbles experience a **shear-gradient lift force**, which attracts them to the viscous sublayer in the channel.
3. This leads to large surface tension stresses near the channel wall, **reducing the budget for viscous stress**, and increasing drag.
4. Surfactants **stabilise the droplet distribution** at intermediate sizes

Thank you! Questions?

I. Cannon, D. Izbassarov, O. Tammisola, L. Brandt, and **M. E. Rosti**, *The effect of droplet coalescence on drag in turbulent channel flows*, Physics of Fluids, 33.8, p85112, Aug. 2021, 10.1063/5.0058632.